

Martin Birkenbaum

SENIOR SOFTWARE ENGINEER

martin.birkenbaum@initech.example.com · [+12065550192](tel:+12065550192) · github.com/mbirkenbaum · [Seattle, WA \(US\)](#) · mbirkenbaum@GitHub

Senior Software Engineer with 8+ years of experience designing, building, and deploying scalable backend services, microservices, and AI infrastructure. Proven ability to develop reliable, large-scale platforms, data pipelines, and GitOps-driven automation systems to support modern AI products. Passionate about testing, model integration, and efficient developer toolchains.

SKILLS

Languages:	Python, TypeScript, JavaScript, SQL, HTML/CSS
Technologies & Frameworks:	Docker, Kubernetes, Git, GitOps, PostgreSQL, pytest, ML Systems, AI Infrastructure, Data Pipelines, Model Inference, LLMs, Distributed Systems, Cloud Environments, Observability, Generative AI, MLOps

WORK EXPERIENCE

Senior Software Engineer at **Initech Corporation** 2021-03-01 – 2026-07-01

Led architecture and development of high-throughput microservices and GitOps-driven infrastructure pipelines.

- Designed and implemented Python-based event-driven microservices and data pipelines, processing 10M+ daily events to support large-scale applications and potential ML inference.
- Standardized containerized deployments utilizing Docker, Kubernetes, and GitOps workflows, enhancing production engineering practices and deployment of complex systems.
- Mentored 6 junior/mid-level engineers, establishing robust code styling and testing standards crucial for reliable system development.
- Developed custom internal toolchains, reducing CI pipeline execution times by 40% and improving operational efficiency for deployment and evaluation workflows.

Software Engineer at **Hooli Inc.** 2018-06-01 – 2021-02-01

Backend engineer focused on building RESTful APIs, developer tooling, and automated deployment integrations.

- Developed robust REST APIs in Python/Django and TypeScript/Node.js, crucial for scalable backend services and system integration.
- Integrated third-party APIs and automated webhook processors, establishing efficient event-driven systems and data ingress for diverse platforms.
- Implemented a comprehensive suite of unit and integration tests, raising coverage from 60% to 92% and ensuring high reliability and performance.

PROJECTS

JobGitOps (Python, Nix, GitHub Actions) 2026-06-01 – 2026-07-01

A GitOps-driven job application and tracking system utilizing GitHub Issues, Projects, and Actions.

- Developed python-jobsby scraper integration and two-pass LLM fit triage
- Designed reproducible local environments via Nix and devenv

EDUCATION

Bachelor of Science in Computer Science at **University of Washington**

2014-09-01 – 2018-06-01

GPA: 3.8